Banking Transaction Analysis - Simple File Review

This is a simple review of every project file: who owns it, what it does, and the main functions.
Roles are based on docs/team_roles.md.

Member 1 - Backend + Integration

.streamlit/config.toml

Does:
Configuration/text file.
Contains: [theme]; base = "light"; primaryColor = "#0F766E"; backgroundColor = "#F4F7FB"

backend/__init__.py

Does:
Python module.

backend/account_service.py

Does:
Account CRUD (MEMBER 1).
Main functions: create_account, get_account, get_all_accounts, get_accounts_by_customer, count_accounts, search_accounts, update_account, deactivate_account, activate_account, count_transactions, delete_account, _transactions_snapshot, get_deleted_accounts, restore_account, purge_deleted_account

backend/balance_service.py

Does:
Balance calculation (MEMBER 1).
Main functions: calculate_balance, get_account_balance, recalculate_balance_in_cursor, get_total_balance, find_balance_mismatches

backend/customer_service.py

Does:
Customer CRUD (MEMBER 1).
Main functions: create_customer, get_customer, get_all_customers, count_customers, search_customers, update_customer, delete_customer

backend/db_connection.py

Does:
Database connection helpers (MEMBER 1).
Main functions: _translate_error, get_connection, get_cursor, fetch_all, fetch_one, execute, check_connection

backend/transaction_service.py

Does:
Transaction operations: deposit, withdrawal, CRUD, search and filters (MEMBER 1).
Main functions: apply_transaction, add_transaction, deposit, withdraw, get_transaction, get_transactions, search_transactions, update_transaction, delete_transaction, get_categories, get_date_range

requirements.txt

Does:

Configuration/text file.
Contains: streamlit>=1.36.0; pandas>=2.0.0; numpy>=1.24.0; plotly>=5.18.0

run.bat
Does:
Configuration/text file.
Contains: @echo off; REM ==; REM Banking Transaction Analysis - launcher; REM Double-click this file to start the dashboard.

tests/test_transactions.py
Does:
Transaction logic tests: deposit, withdrawal, balance rules (needs MySQL).
Main functions: test_opening_balance_is_a_deposit, test_deposit_increases_balance, test_withdrawal_decreases_balance, test_insufficient_balance_rejected, test_invalid_amount_rejected, test_invalid_type_account_and_date_rejected, test_inactive_account_rejected, test_balance_calculation_matches_stored_balance, test_update_and_delete_recalculate_balance, test_update_that_makes_balance_negative_is_rejected, test_filter_and_search, test_sql_injection_text_is_harmless

tests/test_validators.py
Does:
Tests for utils/validators.py (no database needed).
Main functions: test_valid_amount_is_rounded_decimal, test_invalid_amounts_rejected, test_transaction_type, test_date_validation, test_phone_and_email, test_name_account_number_and_id, test_account_type_and_status, test_indian_currency_format

utils/__init__.py
Does:
Python module.

utils/helpers.py
Does:
Small helper functions and constants used in many places.
Main functions: format_inr, generate_reference_number, generate_account_number

utils/validators.py
Does:
Input validators (shared by backend, tests and dashboard).
Main functions: validate_id, validate_name, validate_phone, validate_email, validate_address, validate_account_number, validate_account_type, validate_account_status, validate_transaction_type, validate_amount, validate_date, validate_category, validate_description

Member 2 - Database

database/README.md
Does:
Markdown documentation.
Sections: Database (Member 2); Run it in SQLyog; Tables

database/queries.sql
Does:

SQL script with 13 statements.
Contains: -- ==; -- 1. All customers with the number of accounts they own; -- 2. Accounts with owner name and balance; -- 3. Latest 20 transactions; -- 4. Total deposits and withdrawals; -- 5. Balance of each account calculated from its transactions; -- 6. Monthly expenses; -- 7. Category-wise expenses

database/sample_data.sql
Does:
SQL script with 6 statements.
Contains: -- ==; INSERT INTO customers (name, phone, email, address) VALUES; -- balance starts at 0; it is recalculated from transactions at the bottom; INSERT INTO transactions (account_id, transaction_date, transaction_ty; -- Recalculate every account balance from its transactions

database/schema.sql
Does:
SQL script with 10 statements.
Contains: -- ==; USE banking_db; DROP TABLE IF EXISTS deleted_accounts; DROP TABLE IF EXISTS transactions; DROP TABLE IF EXISTS accounts; DROP TABLE IF EXISTS customers; -- --; -- ----------

tests/test_database.py
Does:
Database tests: connection, tables and customer/account CRUD (needs MySQL).
Main functions: test_database_connection, test_tables_exist, test_customer_create_read_update_search_delete, test_duplicate_customer_rejected, test_invalid_customer_input_rejected, test_account_creation_and_duplicate_number, test_account_for_missing_customer_rejected, test_deactivate_account_and_customer_delete_blocked, test_delete_account_requires_history_confirmation, test_deleted_account_is_archived_and_can_be_restored, test_purge_deleted_account_removes_it_permanently

Member 3 - Data Analysis

analysis/__init__.py
Does:
Python module.

analysis/monthly_analysis.py
Does:
Month-wise analysis (MEMBER 3). Months are shown as 'YYYY-MM'.
Main functions: _with_month, _monthly_total, monthly_expenses, monthly_income, monthly_summary

analysis/report_generator.py
Does:
Build report tables and CSV bytes for download (MEMBER 3).
Main functions: transaction_report, monthly_spending_report, category_spending_report, unusual_transaction_report, build_report, to_csv_bytes, report_file_name

analysis/spending_analysis.py

Does:
Category-wise spending and income vs expense (MEMBER 3).
Main functions: category_expenses, top_spending_categories, income_vs_expense, account_spending

analysis/transaction_analysis.py
Does:
Core transaction analysis with Pandas / NumPy (MEMBER 3).
Main functions: prepare_dataframe, load_transactions, total_deposits, total_withdrawals, calculate_balance, average_transaction, largest_transaction, transaction_statistics

analysis/unusual_transactions.py
Does:
Unusual transaction detection (MEMBER 3).
Main functions: _rule_high_vs_normal, _rule_high_amount, _rule_frequency, unusual_transactions

tests/test_analysis.py
Does:
Analysis tests with a small hand-made DataFrame (no database needed).
Main functions: make_df, make_history_df, test_totals_and_balance, test_average_and_largest, test_statistics, test_monthly_analysis, test_category_analysis_and_income_vs_expense, test_empty_dataframe_does_not_crash, test_unusual_amount_much_higher_than_normal, test_unusual_amount_threshold_is_configurable, test_unusual_frequency_rule, test_csv_export_is_bytes

Member 4 - UI + Visualization

dashboard/__init__.py
Does:
Python module.

dashboard/app.py
Does:
Streamlit entry point (MEMBER 4). Run from the project root with:

dashboard/assets/style.css
Does:
Stylesheet with 123 CSS rules.
Rules: .brand-name; .lp-hero > *; .section-title; .lp-card:hover; [data-testid="stForm"]; :root

dashboard/auth.py
Does:
Demo authentication helpers for the Banking Transaction Analysis app.
Main functions: is_authenticated, current_user, logout, attempt_login

dashboard/components/__init__.py
Does:
Python module.

dashboard/components/charts.py
Does:
Plotly chart builders (MEMBER 4). Each function takes a DataFrame and returns a figure.
Main functions: _style, _empty_figure, monthly_expense_chart, income_vs_expense_chart, category_pie_chart, category_bar_chart, monthly_net_chart, account_comparison_chart,

amount_histogram

dashboard/components/tables.py
Does:
Shared Streamlit widgets: tables, filters, error handling (MEMBER 4).
Main functions: run_safely, records_to_df, show_table, show_chart, show_money, account_options, transaction_filters, unusual_settings, show_unusual_transactions

dashboard/landing.py
Does:
Public landing page shown before the demo login portal.
Main functions: _stats_html, _feature_card, render

dashboard/login.py
Does:
Separate demo login portal for the Banking Transaction Analysis app.
Main functions: _tame_password_manager, render

dashboard/pages/__init__.py
Does:
Python module.

dashboard/pages/accounts.py
Does:
Account management page (MEMBER 4).
Main functions: render

dashboard/pages/analysis.py
Does:
Analysis page: statistics, monthly, category, income vs expense, unusual (MEMBER 4).
Main functions: render

dashboard/pages/customers.py
Does:
Customer management page (MEMBER 4).
Main functions: _customer_label, render

dashboard/pages/dashboard.py
Does:
Main dashboard page: KPI cards, charts, recent transactions, alerts (MEMBER 4).
Main functions: render

dashboard/pages/reports.py
Does:
Report page: preview and CSV download (MEMBER 4).
Main functions: render

dashboard/pages/transactions.py
Does:
Transaction management page: view/search/filter, add, edit, delete (MEMBER 4).
Main functions: _view_tab, _add_tab, _edit_tab, render

dashboard/theme.py

Does:
Theme helpers: loads the custom CSS and renders small HTML components.
Main functions: apply_theme, page_header, section_title, kpi_card

Shared (all members)

.gitignore
Does:
Configuration/text file.
Contains: # secrets; .env; # python; venv/

README.md
Does:
Markdown documentation.
Sections: Banking Transaction Analysis System; Features; Technology stack; Architecture; Database structure; Folder structure; Team responsibilities; Installation

docs/architecture.md
Does:
Markdown documentation.
Sections: Architecture; Rules; Key design decisions

docs/database_design.md
Does:
Markdown documentation.
Sections: Database design (banking_db); Entities and relationships; customers; accounts; transactions; Normalization

docs/file_function_review.pdf
Does:
Configuration/text file.
Contains: %PDF-1.4; 6 0 obj; << /Length 2560 >>; stream

docs/project_workflow.md
Does:
Markdown documentation.
Sections: Project workflow (Git + GitHub); Branches; One-time setup (Member 1); Everyone else; Keeping your branch up to date; How Member 1 integrates the branches

docs/team_roles.md
Does:
Markdown documentation.
Sections: Team roles; Interfaces between members

tests/__init__.py
Does:
Python module.

tests/conftest.py
Does:
Shared pytest setup. Database tests are skipped automatically when MySQL is not reachable.
Main functions: unique_phone, unique_email, require_database, customer_id, account_id

Quick role summary

- Member 1 - Backend + Integration: database connection, CRUD services, balance logic, utils, requirements, launcher.
- Member 2 - Database: schema, sample data, SQL queries, database README.
- Member 3 - Data Analysis: Pandas/NumPy analysis, unusual transaction rules, reports, analysis tests.
- Member 4 - UI + Visualization: Streamlit pages, Plotly charts, filters, tables, login/landing.
- Shared: docs, README, tests helpers.